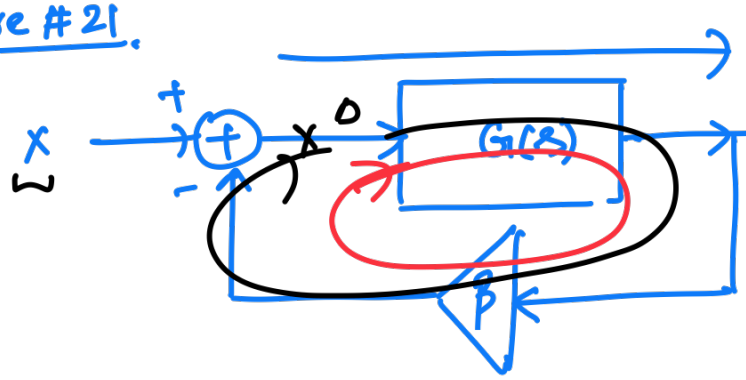
# Lecture #21




$$\frac{Y}{X} = \frac{\frac{1}{\beta}\,L(s)}{1+L(s)}$$

$$L(s) = \beta G(s)$$

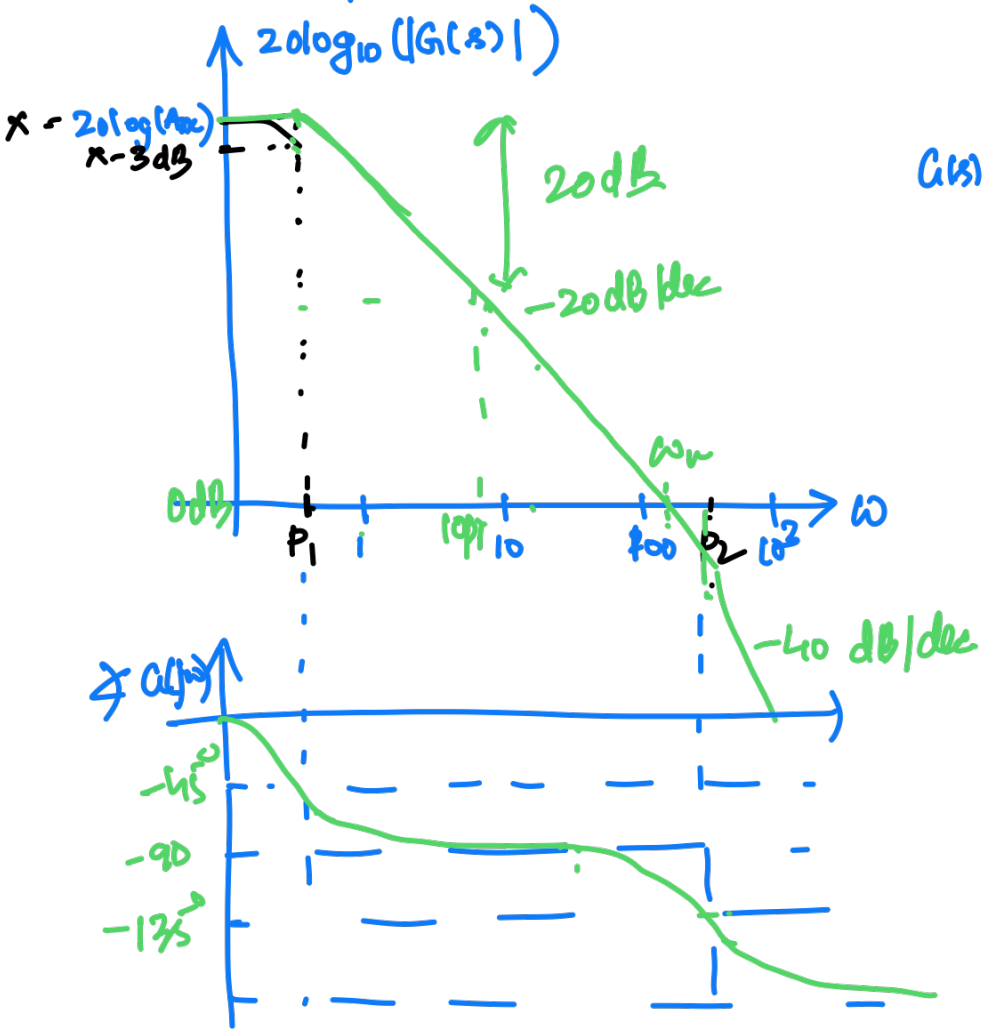
$$G(s) = \frac{A_{DC}}{\left(1+\frac{s}{p_1}\right)\left(1+\frac{s}{p_2}\right)} \quad \text{(forward path gain)}$$

$$= \frac{A_{DC}}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1}$$

$p_1, p_2 > 0$

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{\omega}{p_1}\right) - \tan^{-1}\left(\frac{\omega}{p_2}\right)$$

$H(s) = \beta$ (feedback gain)




$G(s) \xrightarrow{s=j\omega} G(j\omega)$

$G(0) = A_{DC}$

$$G(jp_1) = \frac{A_{DC}}{\left(1+j\frac{p_1}{p_1}\right)\underbrace{\left(1+j\frac{p_1}{p_2}\right)}_{\frac{p_1}{p_2}\ll 1}}$$

$1+j$

$p_1 < \omega < p_2$

$$G(j\omega) \simeq \frac{A_{DC}}{\left(j\frac{\omega}{p_1}\right)1} \checkmark$$

$$|G(j\omega)| \simeq \frac{A_{DC}p_1}{\omega_1}$$

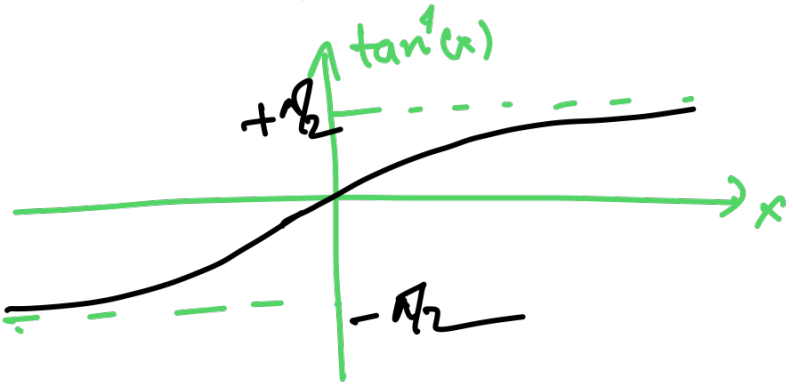
$$|G(j\omega_1)| = \frac{A_{DC}p_1}{10\omega_1}$$

$$|A(j\omega_u)| = \frac{A_{DC}}{\sqrt{1+\frac{\omega_u^2}{p_1^2}}\sqrt{1+\frac{\omega_u^2}{p_2^2}}} = 1$$

$$|A(j\omega)| \underset{\omega \gg p_2}{=} \frac{A_{DC}}{j\frac{\omega}{p_1} \times j\frac{\omega}{p_2}}$$

$$\angle A(jp_1) = -\tan^{-1}\left(\frac{p_1}{p_1}\right) - \underbrace{\tan^{-1}\left(\frac{p_1}{p_2}\right)}_{\simeq 0}$$

$$= -45^\circ$$



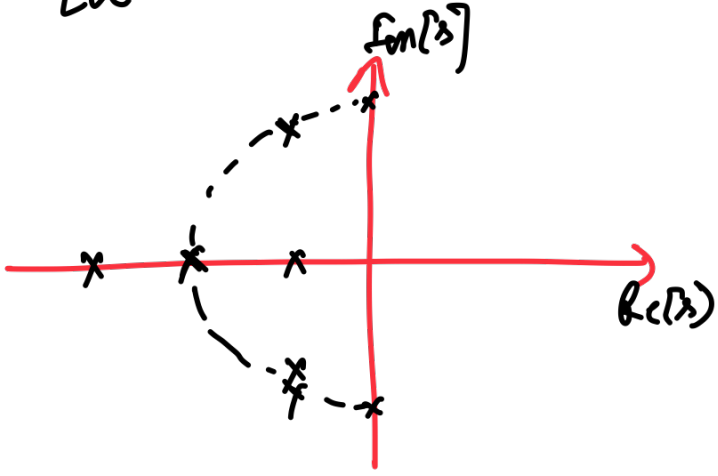

$$\frac{Y}{X} = \frac{1}{\beta}\frac{L\!A}{1+L\!A} = \frac{1}{\beta}\frac{1}{1+\frac{1}{L\!A}}$$

$$D(s) = 1 + L\!A$$

$$= 1 + \beta \cdot A(s)$$

$$= 1 + \beta \frac{A_{DC}}{(1+s/p_1)(1+s/p_2)}$$

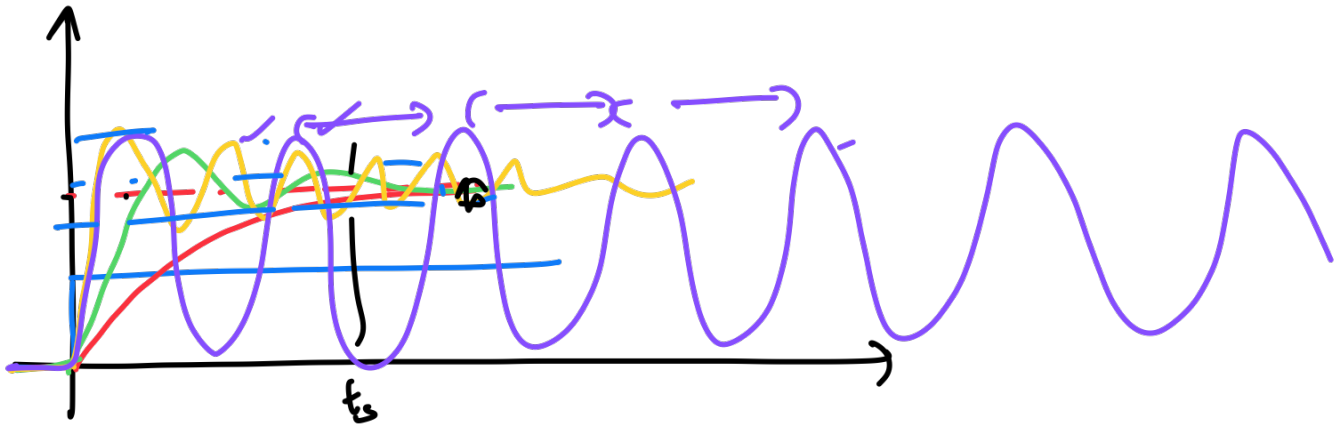
$$= \frac{(1+s/p_1)(1+s/p_2) + \beta A_{DC}}{(\quad)}$$




$$D'(s) = (1+s/p_1)(1+s/p_2) + \beta A_{DC}$$

"Nyquist Stability Criterion" → "loop gain & phase margin in all pole systems."

- if $\phi_m$ (phase margin) $< 0$ $\Rightarrow$ R.H.P poles.




$x(t) \rightarrow [1, t_d] \rightarrow y \rightarrow x(t-t_d)$

$\frac{Y}{X} = e^{-st_d}$

$e^{-j\omega t_d} = \cos(j\omega t_d) - j\sin(\omega t_d)$

$LL(s) = \frac{B\ Arc}{(1+s/p_1)(1+s/p_2)}\ e^{-st_d}$

$\angle e^{-j\omega t_d} =$

$\angle LL(j\omega_n) =$ $-\phi$

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- Stable system (L.H.P poles in closed)
- DC gain margin, phase margin.

$\left.\begin{matrix} 100 \\ 200 \end{matrix}\right\}$ , $\left.\begin{matrix} 70^\circ \\ 45^\circ \end{matrix}\right\}$

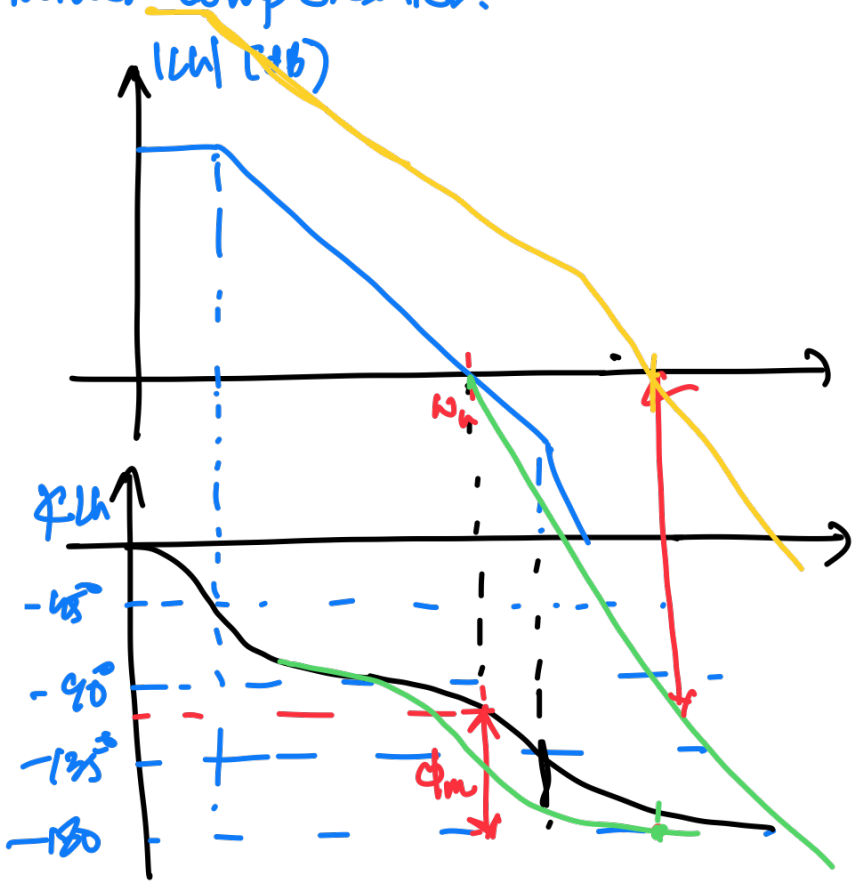
$$A(s) = \frac{A_{DC}}{(1+s/p_1)(1+s/p_2)}$$

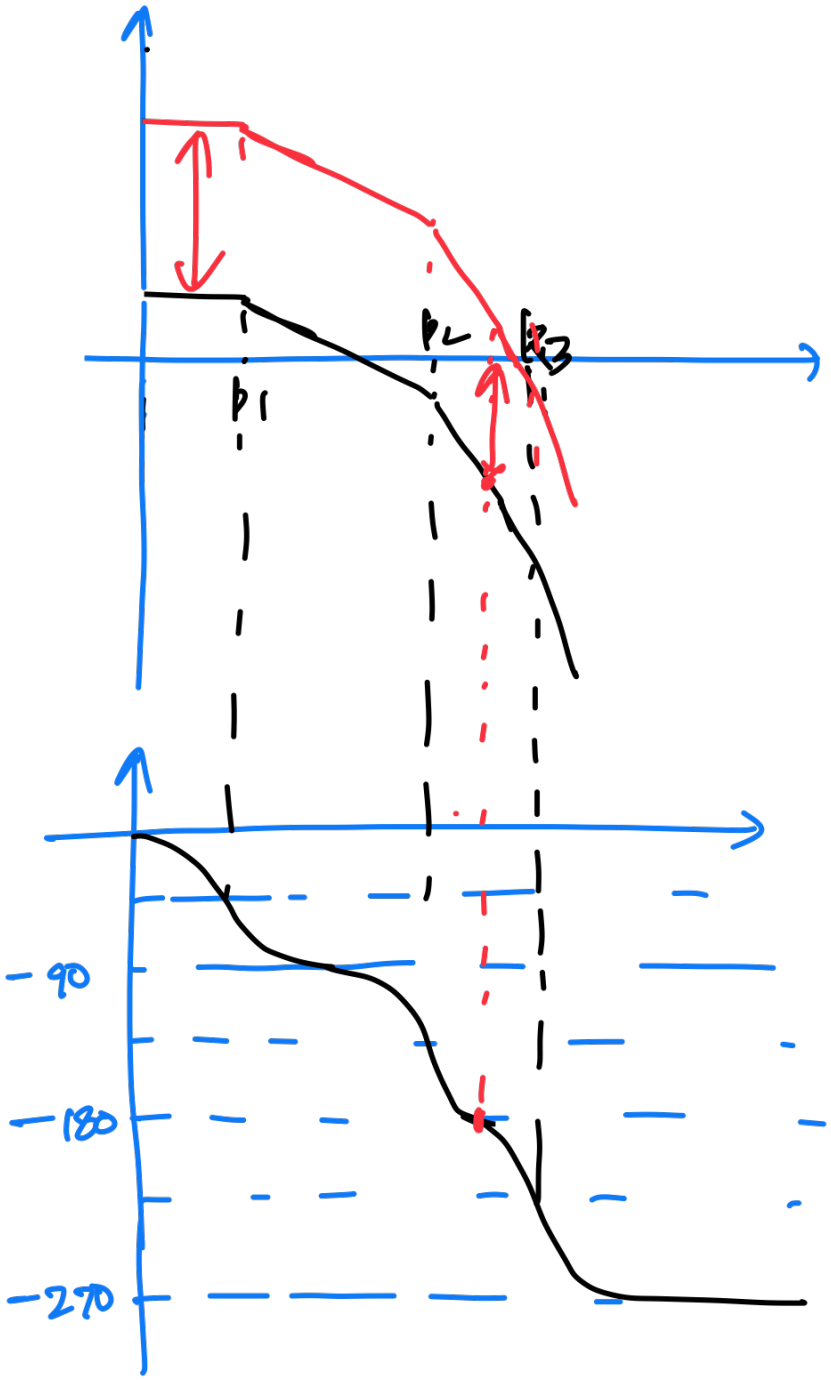
$\beta$ = feed back gain

$$L_A = \beta A(s)$$

↳ Dominant pole compensation. ( large cap. value)

↳ Miller compensation.




-90
-180
-270